Segment focused portfolio optimization based on machine learning predictions

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Motivation

- Noisy and high-frequency data generated on the stock exchange markets have been point of interest for decades.
- Being able to trust-worthy predict day-to-day changes of stock assets would be extremely beneficial for the industry – more informed decision in portfolio management, strategy evaluations, etc.
- There are quite a few studies involing both classification and regression machine learning (ML) approaches to predict developments on the market; however, many of them apply ML techniques only on very limited number of stocks [1, 2] or on stock indices [3].
- What is more is lack of studies applying prediction modelling in portfolio strategy optimization.

- This study evaluates prediction capabilities of selected ML algorithms and portfolio optimization framework on segment of **Information technology** (IT) among 500 biggest US companies by market cap during the period between September 2017 and January 2026.

Modeling pipeline

- We adopted **rolling window** framework for next-day closing price prediction of stocks. Data from trading days $t-q, t-(q-1), \ldots, t-1$ are employed as predictors for the closing price on day t.
- This methodology ensures short-term resilience of models to market shifts and unexpected events.

- Among the included predictor data, historical developments of the IT stocks, **stock indices** (S&P 500, NASDAQ Composite), **commodities** (crude oil, gold), **major FOREX pairs** and **US Treasury Bonds** were included.
- The whole analyzed period (between 2017 and 2026) was split into training-validation (September 2017 to December 2022) and testing interval (January 2023 to January 2026).
- For each rolling window training period and for each stock (out of the total number of 65), ML models were trained and tuned. At the end, the most convenient (in terms of evaluation metrics) general hyperparameter setting was chosen.

Prediction results & Benchmarking

- Several studies lack qualified benchmarking of obtained predictions to some naïve or low-effort solutions. In accordance with Fama [4], we employed benchmarking naïve prediction, assuming price on next day is equal to the yesterday's one.

Dominance corresponds to the fraction of predictions in average, when tuned model was more precise than naïve benchmark. Results of predictive hyper-tuned predictive models (on both training-validation and test samples – results for testing are in corresponding brackets) are as follows:

Model Type	MAPE (%)	Dominance (%)
Naïve benchmark	1.69 (1.73)	
ARIMA	1.90 (1.92)	58.25 (56.68)
Ridge (Stock-wise mode)	1.79 (1.79)	45.15 (45.94)
Ridge (General model)	1.79 (1.80)	45.08 (45.88)
Lasso (General model)	2.30 (2.24)	36.16 (38.83)
Lasso (Stock-wise model)	2.27 (2.31)	35.81 (37.10)

Out-performing naïve benchmark is extremely difficult; however, ARIMA out-performed it in dominance.
Results between training-validation and testing interval are stable – possibly no overfitting.

Model-backed portfolio optimization

- Obtained models are capable of reinforcing portfolio optimization frameworks (models are used to define expected returns). In the case of this study, modified **Markowitz mean-variance portfolio** was chosen [5].
- Employed modified version takes into account **transaction costs**.
  Let $\mathbf{w}_t, \mathbf{w}_{t-1}$ be assets weights in days $t, t-1$ respectively; $\boldsymbol{\mu}_t$ expected returns in day t; $\boldsymbol{\Sigma}$ covariance matrix and λ_R, λ_T hyperparameters of risk-aversion and transaction unit cost respectively; then capital allocation optimization is given as:

$$\max_{\mathbf{w}_t} \quad \mathbf{w}_t^\top \boldsymbol{\mu}_t - \lambda_R \mathbf{w}_t^\top \boldsymbol{\Sigma} \mathbf{w}_t - \lambda_T \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1$$

$$\text{Subject to:} \quad \mathbf{w}_t \geq \mathbf{0}_a, \quad \mathbf{1}_a^\top \mathbf{w}_t = 1.$$

Tuned portfolio models evaluations, on the interval between January 2023 and January 2026 (Testing interval in corresponfing brackets), are given:

Portfolio Type	Annual Return (%)	Annual Volatility (%)
Naïve	22.38 (16.48)	17.01 (25.20)
ARIMA	22.25 (15.67)	16.94 (25.48)
Ridge	21.56 (15.30)	16.75 (25.24)
Lasso	20.15 (17.07)	16.80 (25.26)
S&P 500	7.69 (8.35)	21.75 (23.10)
NASDAQ	9.19 (18.51)	25.52 (25.15)

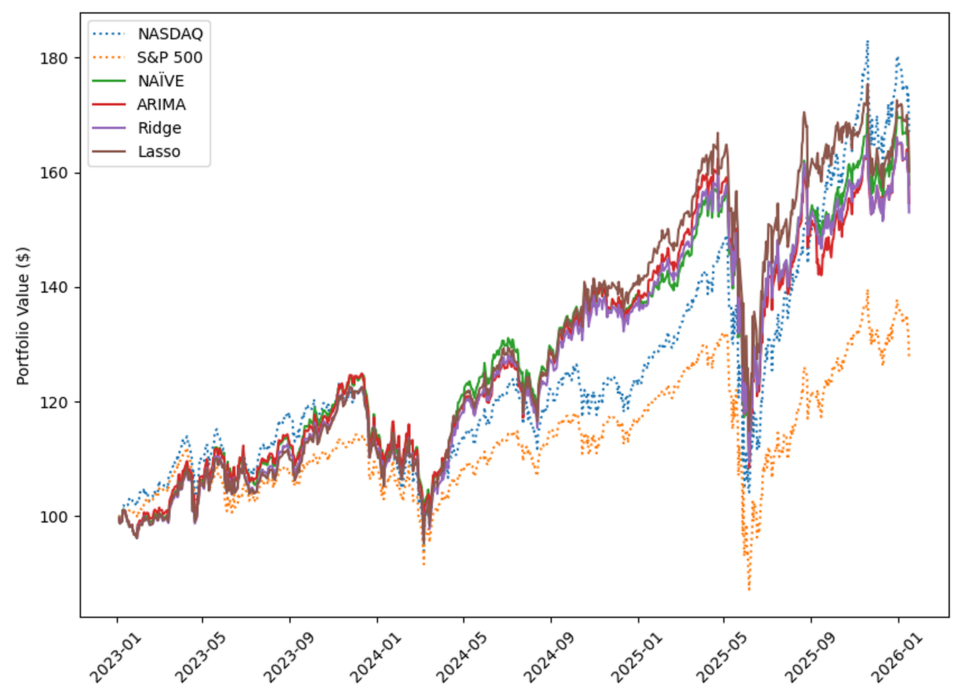

Figure 1: Performance of a $100\ \$$ investment in proposed model and indices (Testing interval).

Conclusions & Discussion

- Time-dependence aware model (e.g. ARIMA) are capable of delivering comparatively better results, rather than conventional ML solutions.
- In order to build resilient investment strategy, extremely precise predictions of returns are not necessarily essential.

- During market phase of neither bull or bear (Training-validation interval), proposed approach involving prediction models is probably capable of delivering above-average results; however, during bull periods (2023 – 2026, AI surge), strategy could not compete with NASDAQ benchmark. Even though, general market (S&P 500) was beaten (in terms of annual return).
- Resulting data-pipeline is universal, capable of analyzing any set of stocks, segment, etc.

KEY REFERENCES

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GITHUB REPOSITORY

Original research done for bachelor thesis. Further sources are saved as a Git repository.

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